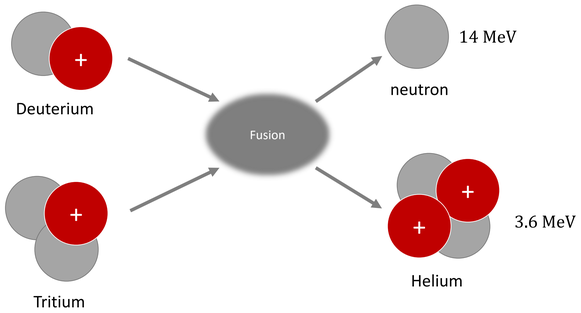
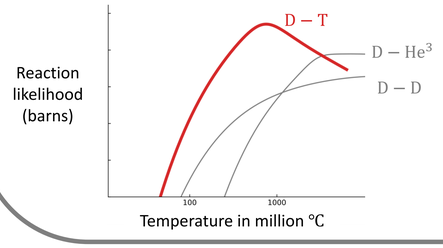
ARC

Affordable Robust Compact tokamak

Concept from:
Sorbom et al (2015), Fus. Eng. Design,
DOI: 10.1016/j.fusengdes.2015.07.008

What is Deuterium-Tritium fusion?



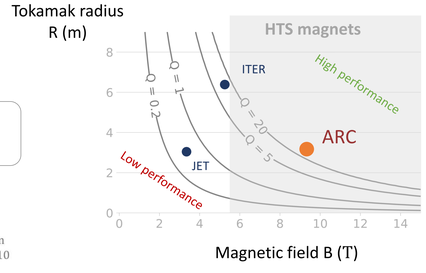


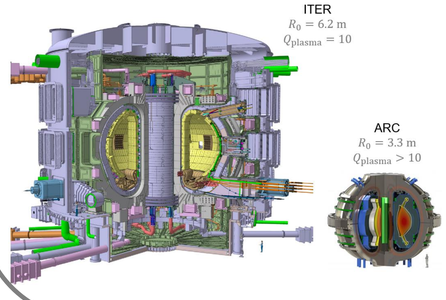
← Why do we need Deuterium and Tritium?

High Temperature Superconducting Magnet

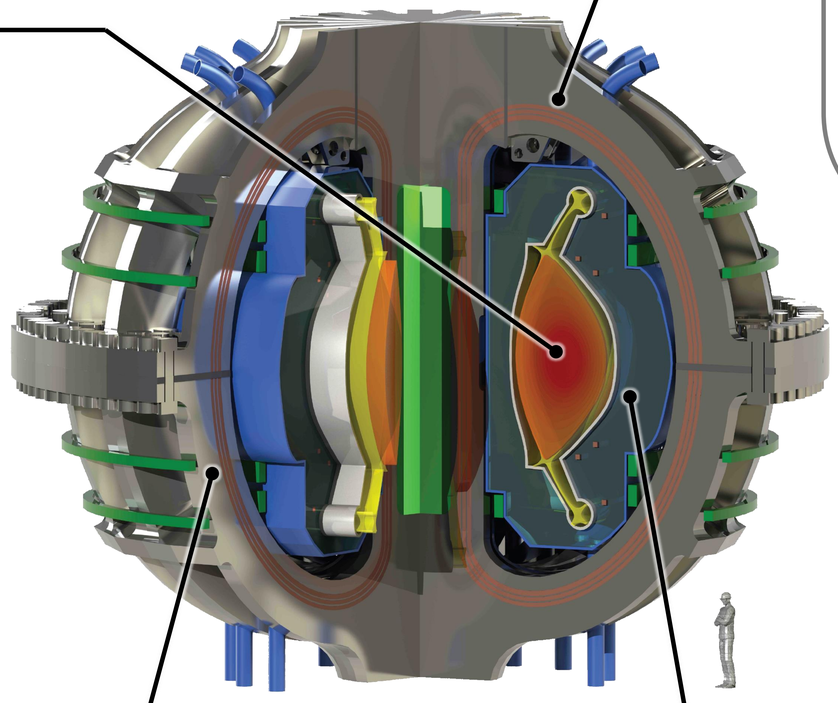
Increasing the magnetic field greatly improves fusion performance

$$Q_{\text{plasma}} = \frac{\text{fusion power}}{\text{heating power}} \propto R^3 \boldsymbol{B}^5$$





Using HTS magnets allows tokamaks to be more compact and cheaper



Major radius	3.3 m
TBR	1.1
Fusion power	525 MW
Multiplication factor Q	> 10
Magnetic field	9.2 T

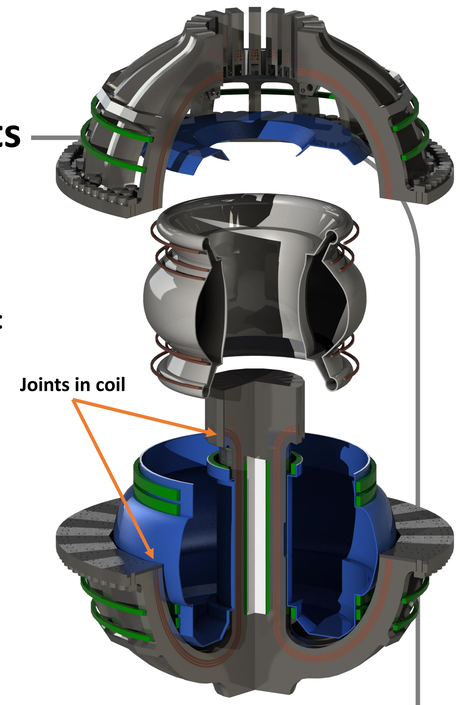
Demountable magnets

Using **H**igh **T**emperature **S**uperconductors enables joints:

☑ Magnets can be mounted/demounted many times

☑ Easier maintenance from the top

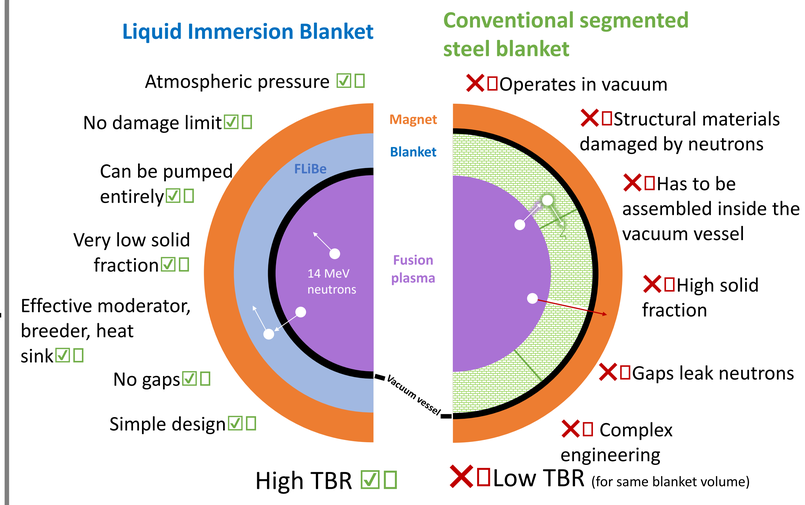
☑ Vacuum vessel can be replaced and is no longer a lifetime component



ARC's assembly video

Liquid Immersion Blanket

⚛ Breeding tritium ? Shield from neutrons 🔥 Extract heat



Why Lithium?

$$\mathrm{D} + \mathrm{T} \rightarrow \mathrm{He} + \mathrm{n}$$
$$\mathrm{Li} + \mathrm{n} \rightarrow \mathrm{T} + \mathrm{He}$$

Why Beryllium?

$$^{9}\mathrm{Be} + \mathrm{n} \rightarrow {}^{8}\mathrm{Be} + \mathrm{n} + \mathrm{n}$$

$$\textbf{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Theoretical sufficiency → TBR = 1
Losses + start other plants → $\mathbf{TBR \approx 1.1 - 1.2}$